

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
Department of Geosciences and CGISS, Boise State University
Environmental Research Building (ERB) 1104 (lab), 3155 (office)
1910 University Drive
Boise, ID 83725

(208) 426-1416 (office)

(303) 859-3106 (cell)

hpmarshall@boisestate.edu

<https://www.boisestate.edu/earth-cryogars/>

Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Selected Professional Experience

- 08/2018-8/2021 Project Scientist, NASA SnowEx Mission
- 08/2016-07/2018 Co-lead of Field Campaign, NASA SnowEx 2017 Mission
- 08/2022-present Professor, Department of Geosciences, Boise State University

Selected Recent Peer-Reviewed Publications

[h-index=33, i10-index=71, citations=3269]

- 2024 Rudisill, W., Flores, A., **Marshall, H.P.**, Siirila-Woodburn, E., Feldman, D., Rhoades, A., Xu, Z., and Morales, A. (2024). Cold-season precipitation sensitivity to microphysical parameterizations: Hydrologic evaluations leveraging snow lidar datasets. *Journal of Hydrometeorology*, 25(1):143–160
- 2023 Hoppinen, Z., Johnson, J. B., Capelli, A., Anderson, J. H., and **Marshall, H.P.** (2023). Snow attenuation of infrasound signals and wind noise. *Seismological Research Letters*, doi:10.1785/0220230048
- Bonnell, R., McGrath, D., Hedrick, A. R., Trujillo, E., Meehan, T. G., Williams, K., **Marshall, H.P.**, Sexstone, G., Fulton, J., Ronayne, M. J., Fassnacht, S. R., Webb, R. W., and Hale, K. E. (2023). Snowpack relative permittivity and density derived from near-coincident lidar and ground-penetrating radar. *Hydrological Processes*, 37(10):e14996
- Hojatimalekshah, A., Gongora, J., Enterkine, J., Glenn, N. F., Caughlin, T., **Marshall, H.P.**, and Hiemstra, C. A. (2023). Lidar and deep learning reveal forest structural controls on snowpack. *Frontiers in Ecology and the Environment*, 21(1):49–54
- Tarricone, J., Webb, R., **Marshall, H.P.**, Nolin, A., and Meyer, F. (2023). Estimating snow accumulation and ablation with l-band insar. *The Cryosphere*, 17(5):1997–2019